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R18

Course Code: A30005



CMR COLLEGE OF ENGINEERING & TECHNOLOGY
(UGC AUTONOMOUS)

B.Tech II Semester Supplementary Examinations September-2023

Course Name: ODES & MULTIVARIABLE CALCULUS

(Common for all Branches)

Date: 04.09.2023 AN

Time: 3 hours

Max.Marks: 70

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions (Compulsory)

Each question carries TWO marks.

10x2=20M

1. Find the integrating factor of $y^2 dx + (x^2 - xy - y^2) dy = 0$ 2 M
2. Define the order and degree of a Differential equation. 2 M
3. Find the Particular Integral of $(D^2 + 16)y = \cos 4x$ 2 M
4. Find the complimentary function of $(D^2 + 4D + 3)y = 0$ 2 M
5. Evaluate $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz$ 2 M
6. Evaluate $\int_0^1 \int_0^1 xy dx dy$ 2 M
7. If $\phi = x^2 + y^2 + z^2$ then find $\text{grad } \phi$ at $(1, 1, 1)$ 2 M
8. If $\vec{r} = xi + yj + zk$ then find $\nabla \times \vec{r}$ 2 M
9. Give the physical significance of line integral $\int_c \vec{F} \cdot d\vec{r}$ 2 M
10. State Gauss divergence theorem. 2 M

PART-B

Answer the following. Each question carries TEN Marks.

5x10=50M

- 11.A). Solve $(x^2 + y^2)dx - 2xydy = 0$ 10M
- OR**
11. B). A bacterial culture growing exponentially increases from 200 to 500 grams in the period from 6 A.M to 9 A.M. How many grams will be present at noon? 10M
12. A). Solve $(D^2 - 3D + 2)y = x^2 + \cos 3x$ 10M
- OR**
12. B). Solve $(D^2 - 2D + 2)y = e^x \tan x$ by using method of variation of parameters. 10M
13. A). Evaluate the integral by changing the order of integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ 10M
- OR**
13. B). Find the area bounded by the curves $y^2 = 4x$ & $x^2 = 4y$ 10M

(P.T.O..)

14. A). Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point P (1, 2, 3) in the direction of the line PQ where Q = (5, 0, 4). 10M

OR

14. B). Find the angle between the two surfaces $x^2 + y^2 + z^2 = 9$ & $z = x^2 + y^2 - 3$ at the point (2, -1, 2) 10M

15. A). Verify Green's theorem in the plane for $\oint_C (x^2 - xy^3)dx + (y^2 - 2xy)dy$ where C is a square with vertices (0,0), (2,0), (2,2), (0,2). 10M

OR

15. B). Evaluate by stoke theorem $\oint_C (x + y)dx + (2x - z)dy + (y + z)dz$ where C is the boundary of the triangle with vertices (0,0,0), (1,0,0) and (1,1,0) 10M
